synthesis. Execution-guided approaches leverage intermediate execution outcomes to guide program search [86–88]. Another line of research utilizes majority voting to choose candidates based on their execution performance [89, 90]. Additionally, LEVER [91] trains a verifier to distinguish and reject incorrect programs based on execution results. CLAIRIFY [92], on the other hand, generates code for planning chemistry experiments and makes use of a rule-based verifier to iteratively provide error feedback to LLMs. VOYAGER distinguishes itself from these works by integrating environment feedback, execution errors, and self-verification (to assess task success) into an iterative prompting mechanism for embodied control.

6 Conclusion

In this work, we introduce VOYAGER, the first LLM-powered embodied lifelong learning agent, which leverages GPT-4 to explore the world continuously, develop increasingly sophisticated skills, and make new discoveries consistently without human intervention. VOYAGER exhibits superior performance in discovering novel items, unlocking the Minecraft tech tree, traversing diverse terrains, and applying its learned skill library to unseen tasks in a newly instantiated world. VOYAGER serves as a starting point to develop powerful generalist agents without tuning the model parameters.

7 Broader Impacts

Our research is conducted within Minecraft, a safe and harmless 3D video game environment. While VOYAGER is designed to be generally applicable to other domains, such as robotics, its application to physical robots would require additional attention and the implementation of safety constraints by humans to ensure responsible and secure deployment.

8 Acknowledgements

We are extremely grateful to Ziming Zhu, Kaiyu Yang, Rafał Kocielnik, Colin White, Or Sharir, Sahin Lale, De-An Huang, Jean Kossaifi, Yuncong Yang, Charles Zhang, Minchao Huang, and many other colleagues and friends for their helpful feedback and insightful discussions. This work is done during Guanzhi Wang's internship at NVIDIA. Guanzhi Wang is supported by the Kortschak fellowship in Computing and Mathematical Sciences at Caltech.

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